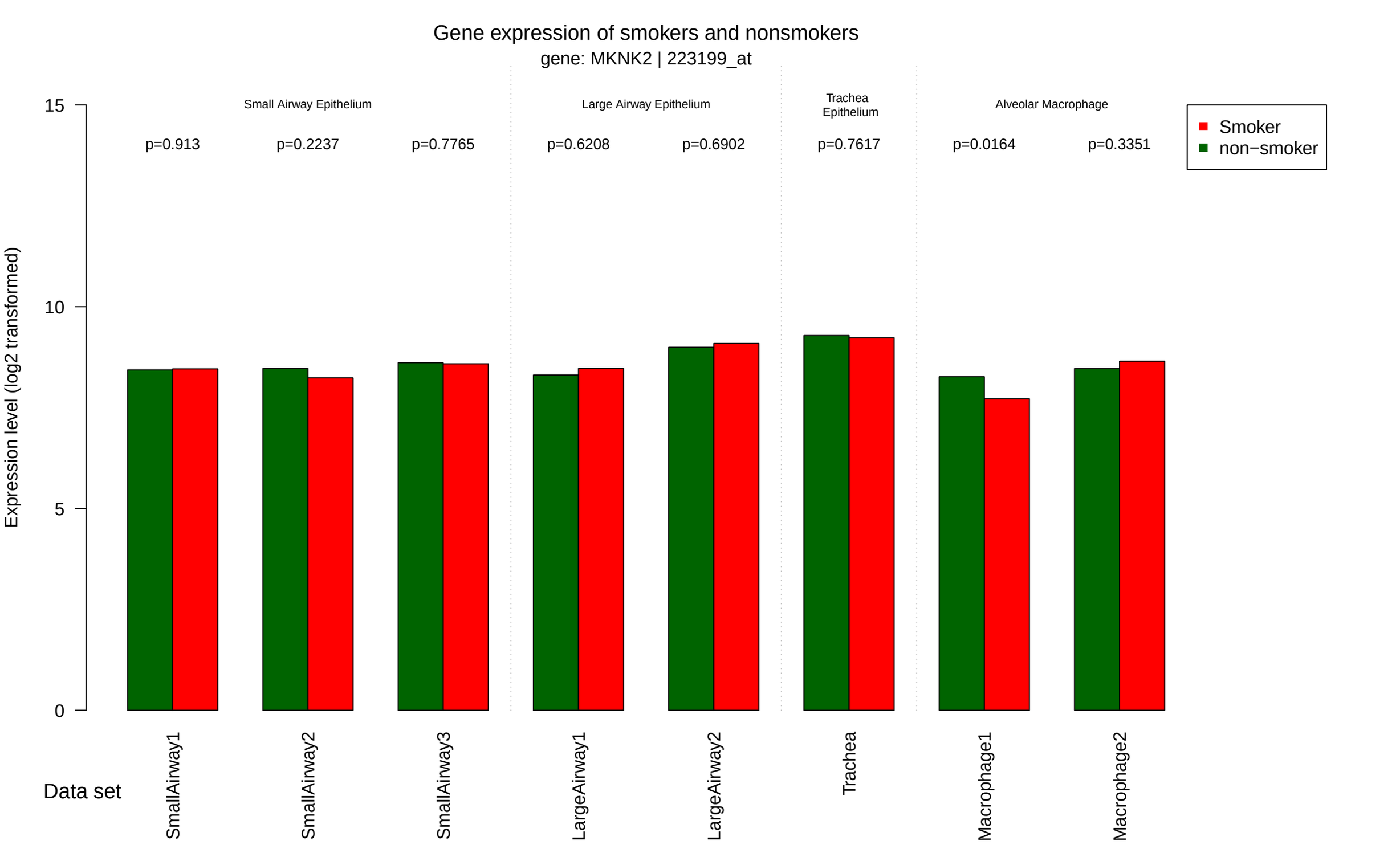
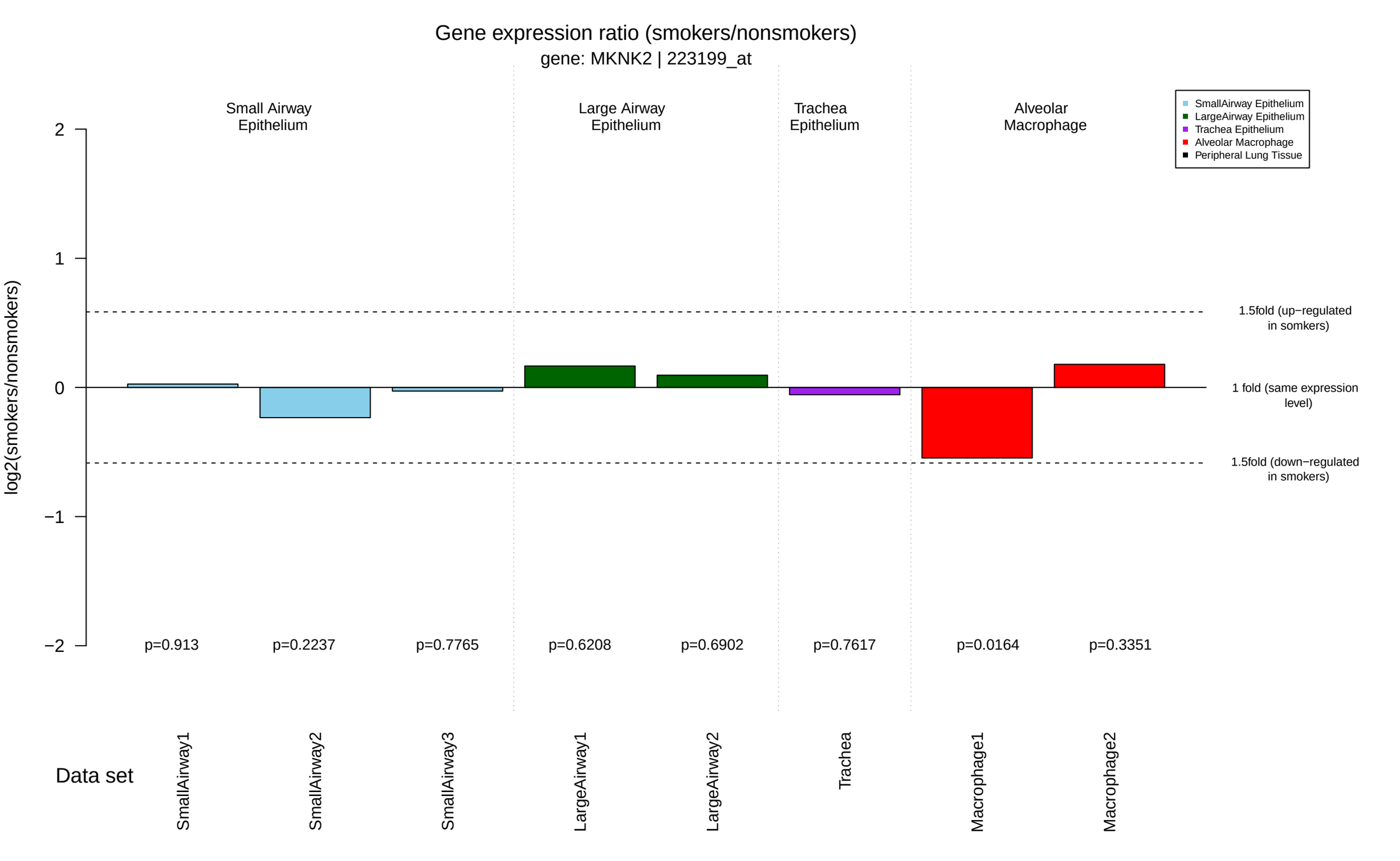
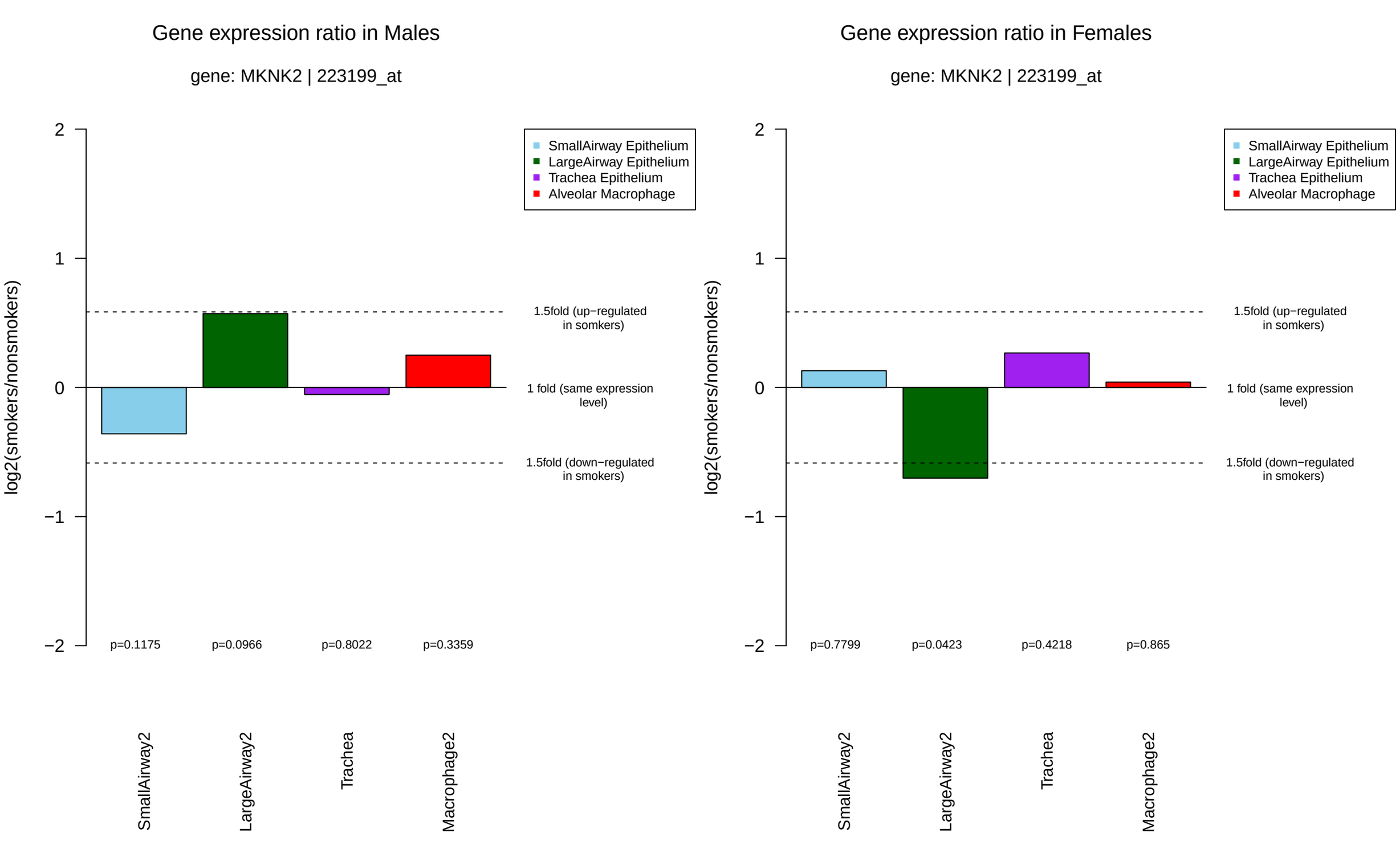
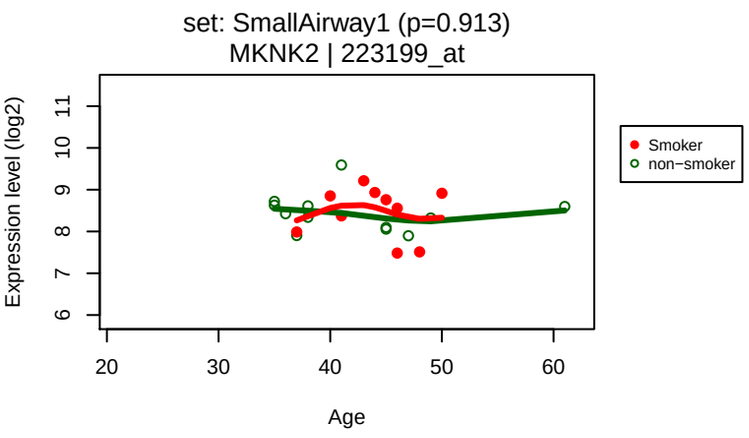
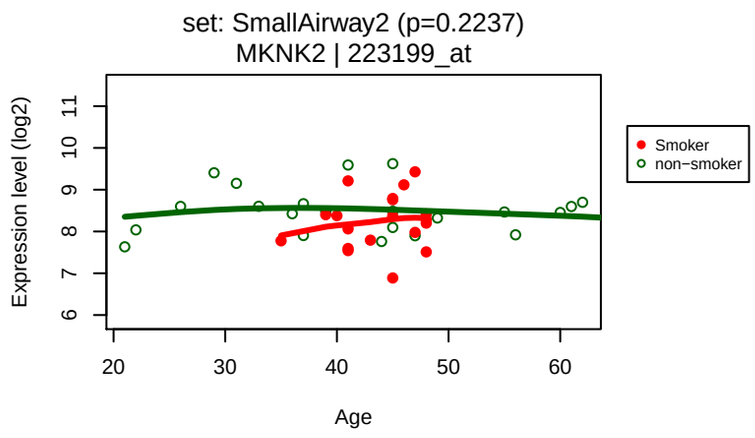
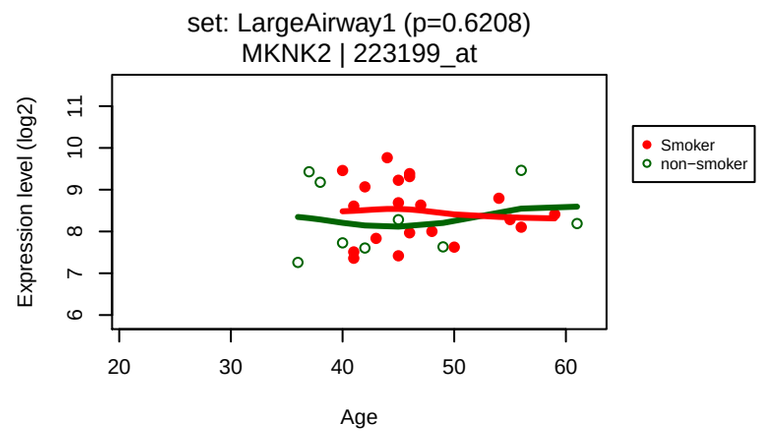

Gene expression of smokers and nonsmokers
gene: MKNK2 | 223199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.913
p=0.2237
p=0.7765
p=0.6208
p=0.6902
p=0.7617
p=0.0164
p=0.3351
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

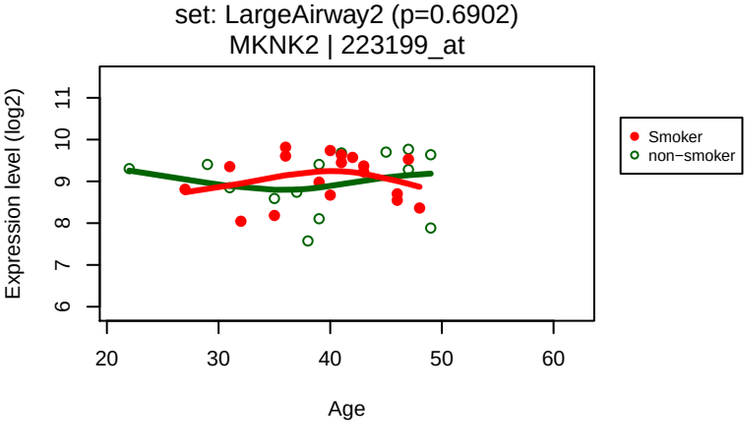

Gene expression ratio (smokers/nonsmokers)
gene: MKNK2 | 223199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.913
p=0.2237
p=0.7765
p=0.6208
p=0.6902
p=0.7617
p=0.0164
p=0.3351
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

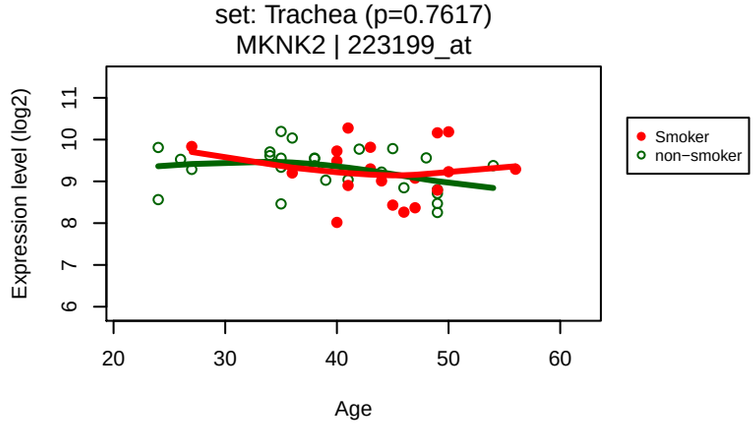

Gene expression ratio in Males
gene: MKNK2 | 223199_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1175
p=0.0966
p=0.8022
p=0.3359
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MKNK2 | 223199_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7799
p=0.0423
p=0.4218
p=0.865
SmallAirway2
LargeAirway2
Trachea
Macrophage2

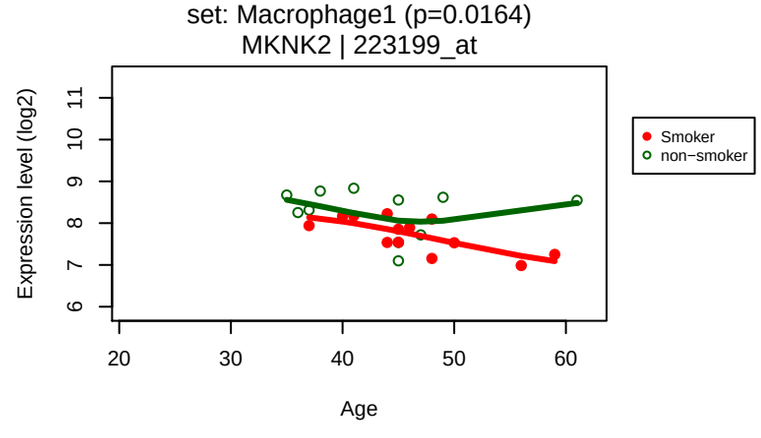

set: SmallAirway1 (p=0.913)
MKNK2 | 223199_at
Expression level (log2)
Age
Smoker
non−smoker

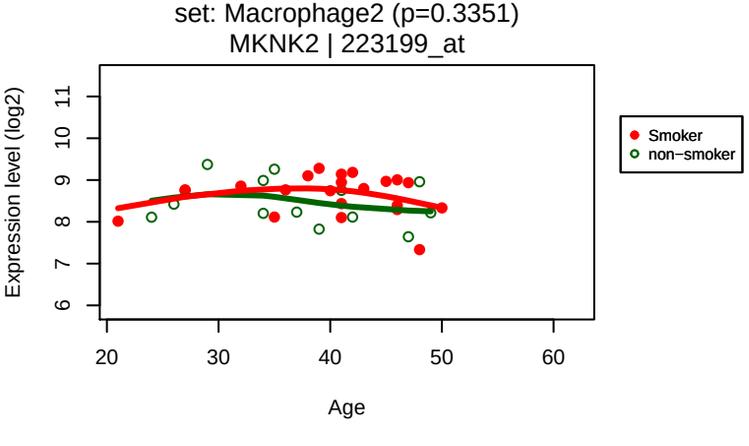

set: SmallAirway2 (p=0.2237)
MKNK2 | 223199_at
Expression level (log2)
Age
Smoker
non−smoker

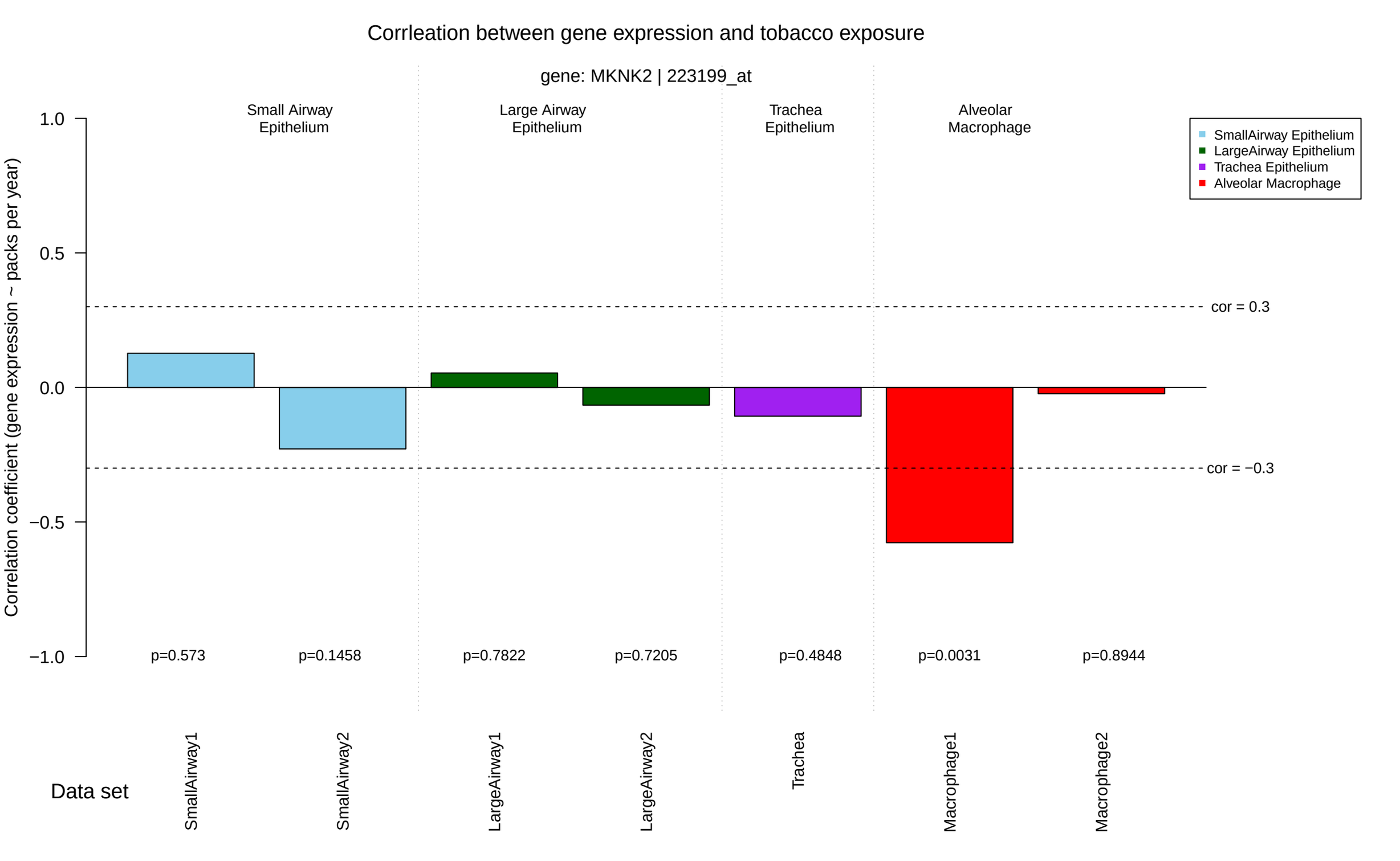

set: LargeAirway1 (p=0.6208)
MKNK2 | 223199_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6902)
MKNK2 | 223199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7617)
MKNK2 | 223199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0164)
MKNK2 | 223199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3351)
MKNK2 | 223199_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MKNK2 | 223199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.573
p=0.1458
p=0.7822
p=0.7205
p=0.4848
p=0.0031
p=0.8944
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2